

A Low-Profile Stacked Dielectric Resonator Antenna With High-Gain and Wide Bandwidth

Y. M. Pan, *Member, IEEE*, and S. Y. Zheng, *Member, IEEE*

Abstract—A wideband, low profile and high gain dielectric resonator antenna is investigated in this letter. The antenna consists of two dielectric layers with different permittivities and it is centrally fed by a rectangular slot. By placing the dielectric layer of low permittivity (2.2) below that of high permittivity (15), the antenna with low profile of $0.1\lambda_0$ can obtain a 10-dB impedance bandwidth of $\sim 40\%$ and an average gain of ~ 9 dBi.

Index Terms—Dielectric resonator antenna, high-gain antenna, low-profile antenna, wideband antenna.

I. INTRODUCTION

SINCE dielectric resonator antenna (DRA) is purely made of dielectric, its loss can be made very small even at mm-wave frequencies. Also, the DRA is a 3-D device, it has higher degree of flexibility and versatility than for the 1-D linear antenna and 2-D planar antenna, allowing for designs to suit various physical or electrical requirements of different communication applications [1]. These attractive features make the DRA to be an ideal candidate for low-loss antenna design. However, the relatively low gain of ~ 5 dBi is a limitation for using DRA in some certain applications. Different methods have been proposed to enhance the gain [2]–[5]. A surface mounted short horn was used to make the gain of a cross DRA higher than 9 dBi [2]. A circular periodic EBG substrate based on the mushroom-like structure was applied for a cylindrical DRA to offer a gain enhancement of 3 dB [3]. The higher-order $TE_{\delta 15}$ mode was excited in a very tall rectangular DRA to obtain a maximum gain of 10 dBi [4]. However, these approaches increase the complexity or height of the DRAs. High profile DRAs are not suitable for applications that have limited space. Also, they are not easier to conform to carriers and may encounter large wind resistance.

Multi-segment DRA or stacked DRA has been proved to be an effective approach to broaden the impedance

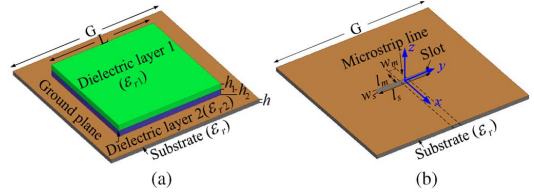


Fig. 1. (a) Configuration of the wideband, low profile and high gain DRA composed of two planar dielectric layers with different permittivities. (b) Feeding circuit of the microstrip-coupled slot.

bandwidth [6]–[9]. It uses a combination of dielectric layers with different permittivities to function as a whole DRA. Impedance bandwidth of up to about 20% or even wider can be achieved by either placing the dielectric layer of low permittivity under or above the layer of high permittivity. Unfortunately, the reported stacked DRAs also generally have high profile and low gain. Although it has been illustrated in [10] that stacking two DRA of very high permittivity (ϵ_r of 38, 80) can achieve a gain improvement of ~ 1.2 dBi, the average gain in the impedance passband is still only ~ 6 dBi.

In this letter, a stacked DRA which consists of two dielectric layers with different permittivities is further investigated. It has been found that by properly choosing the permittivities and dimension ratios, the DRA with low profile of $\sim 0.1\lambda_0$ can easily achieve a quite wide impedance bandwidth of $\sim 40\%$ and a medium average gain of ~ 9 dBi.

II. ANTENNA CONFIGURATION

The antenna configuration is shown in Fig. 1. It consists of a top rectangular dielectric layer with high permittivity ϵ_{r1} and a bottom dielectric layer with low permittivity ϵ_{r2} . The two layers have the same sidelength L but different heights of h_1 and h_2 , respectively. The dimensions are chosen such that $L \gg h_1 + h_2$, therefore the DRA has relatively low profile for a given resonance frequency [11], [12]. The antenna is fed at its center by a microstrip-coupled slot, which is fabricated on a substrate having a permittivity of ϵ_r , a thickness of h and a size of $G \times G$. As shown in Fig. 1(b), the rectangular slot of length l_s and width w_s is fabricated at the top ground plane, and a $50 - \Omega$ microstrip feed-line of width w_m and stub length l_m is etched on the other side of the substrate. Since a y -directed slot is used for excitation, the DRA is x -polarized.

III. SIMULATED AND MEASURED RESULTS

A wideband antenna operating in C band was designed, fabricated, and tested for demonstration. The parameters of the prototype are given by $\epsilon_{r1} = 15$, $h_1 = 2.6$ mm, $\epsilon_{r2} = 2.2$,

Manuscript received March 09, 2015; revised April 24, 2015; accepted May 02, 2015. Date of publication May 05, 2015; date of current version February 04, 2016. This work was supported by the National Natural Science Foundation of China under Grant No. 61302001, the Natural Science Foundation of Guangdong Province under Grant 2014A030313250, and the Open Project of the State Key Laboratory of Millimeter Waves under Grant No. K201512.

Y. M. Pan is with the School of Electronic and Information Engineering, South China University of Technology, Guangzhou 510640, China, and also with the State Key Laboratory of Millimeter Waves, Southeast University, Nanjing 210096, China.

S. Y. Zheng is with the Department of Electronics and Communication Engineering, Sun Yat-sen University, Guangzhou 510275, China (e-mail: ceympan@scut.edu.cn).

Color versions of one or more of the figures in this letter are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/LAWP.2015.2429686

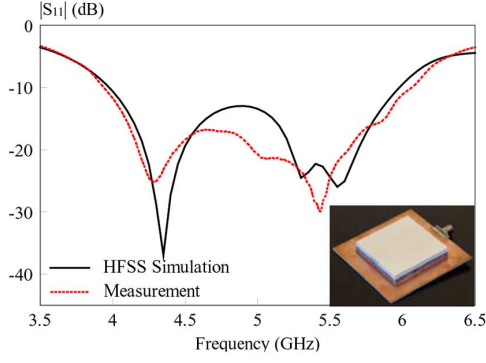


Fig. 2. Simulated and measured reflection coefficients of the wideband, low profile and high gain DRA.

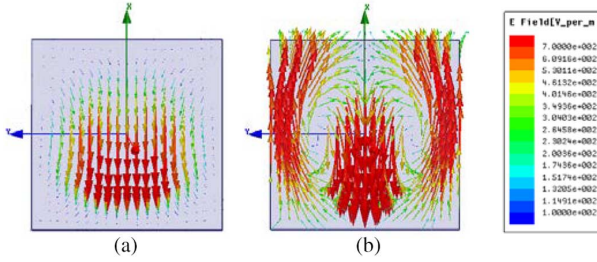


Fig. 3. Simulated resonant E-field inside the DRA. (a) 4.35 GHz. (b) 5.4 GHz.

$h_2 = 3$ mm, $L = 46$ mm, $\epsilon_r = 3.38$, $h = 0.813$ mm, $G = 70$ mm, $l_s = 31$ mm, $w_s = 1.8$ mm, $l_m = 9$ mm, $w_m = 1.88$ mm. In the prototype, the high permittivity layer with $\epsilon_{r1} = 15$ is fabricated by cutting an ECCOSTOCK HiK dielectric material, and the low permittivity layer with $\epsilon_{r2} = 2.2$ is fabricated by etching a commercially available PCB. The two dielectric layers and the feeding substrate are then cemented together by a very thin layer of instant glue, which has negligible effect on the antenna performance. In this letter, the reflection coefficient is measured using an HP8510C network analyzer, whereas the antenna gains and radiation patterns are obtained using a Satimo Starlab System. Fig. 2 shows the simulated and measured reflection coefficients of the prototype, with reasonable agreement between them obtained. The simulated and measured 10-dB impedance bandwidths are 40.4% (3.97-5.98 GHz) and 42.4% (3.94-6.06 GHz), respectively. Two resonant modes can be observed in the passband. Fig. 3 shows the resonant E-field patterns inside the DRA at the lower (4.35 GHz) and upper (5.4 GHz) frequencies, indicating that the first resonance is caused by the TE_{111}^y mode and the second one is due to the TE_{131}^y mode. The middle TE_{121}^y mode cannot be excited because it requires the E-field to be minimum at the location of slot. The measured and simulated far field radiations patterns at frequencies of 4.35 and 5.4 GHz are shown in Fig. 4. As can be seen from the figure, broadside radiation patterns are desirably observed at both frequencies. The copolarized fields in the boresight direction ($\theta = 0^\circ$) are stronger than the cross-polarized fields by near 30 dB. The radiation patterns have also been simulated at other frequencies and very stable results are obtained across the entire passband.

Fig. 5 shows the measured and simulated antenna gains of the proposed antenna. With reference to the figure, the

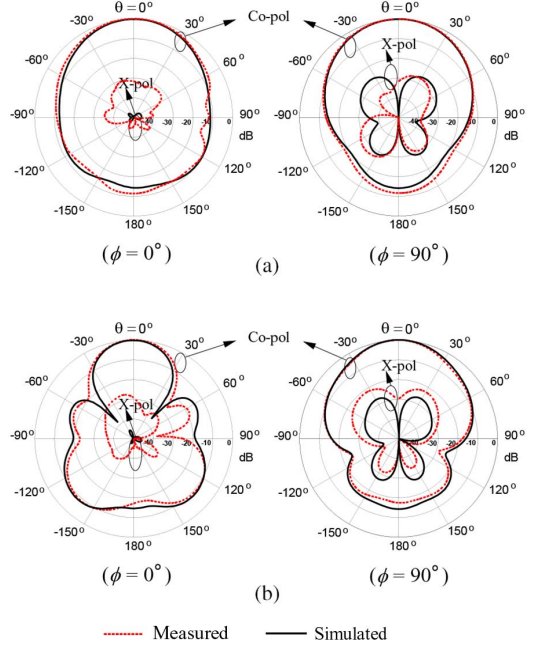


Fig. 4. Simulated and measured radiation patterns of the wideband, low profile and high gain DRA. (a) 4.35 GHz. (b) 5.4 GHz.

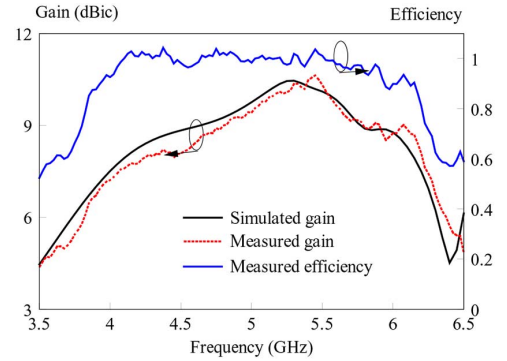


Fig. 5. Gain and efficiency of the wideband, low profile and high gain DRA.

TABLE I
COMPARISON OF THE IMPEDANCE BANDWIDTH, MAXIMUM GAIN, ANTENNA HEIGHT BETWEEN THE PROPOSED DRA AND THE PREVIOUSLY REPORTED DESIGNS

Antennas	Impedance bandwidth	Maximum antenna gain	Antenna height (in terms of λ_g)
Present design	40 %	10.5 dBi	0.267
Design in [2]	21.3 %	10.1 dBi	0.613
Design in [3]	6 %	8.3 dBi	0.562
Design in [4]	7.5 %	10.2 dBi	3.448
Design in [5]	15 %	11.9 dBi	0.643

measured gain is generally lower than the simulated value. This is because the simulated result has not taken into account the losses of the dielectric, connectors, and cables. The maximum gain of 10.5 dBi is found at ~ 5.4 GHz, and the average gain across the impedance passband is ~ 9 dBi. A comparison of the impedance bandwidth, maximum gain and antenna height between the proposed antenna and the previously reported

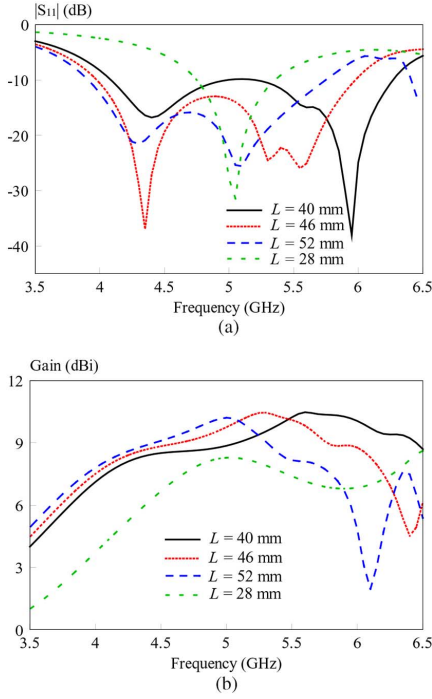


Fig. 6. Simulated reflection coefficients and gains of the wideband, low profile and high gain DRA as a function of frequency for different side-lengths (L) of the two dielectric layers. (a) Reflection coefficients. (b) Antenna gains.

designs are summarized in Table I. From the table, it can be observed that the present antenna possesses satisfying performances. Fig. 5 also shows the measured antenna efficiency, which is as high as 95% at most frequencies.

IV. PARAMETRIC STUDY AND DISCUSSION

A parametric study has been carried out using HFSS. In each case, only one parameter is varied while the others are kept the same as that of the above prototype. Firstly, the effect of the DRA dimensions is studied. Fig. 6 shows the reflection coefficients and gains versus frequency for different sidelengths L of the dielectric layers. It can be seen from Fig. 6(a) that as L increases from 40 to 52 mm, the lower resonance frequency at ~ 4.4 GHz remains almost unchanged. This is expected for a low profile DRA with $L \gg h$, because the resonance is due to the wave bouncing between the top and the bottom faces of the DR [13]. However, the upper resonance frequency decreases significantly from 6 to 5.1 GHz with the increase of L , showing that the upper mode is mainly caused by the resonance along x direction. As can be seen from Fig. 6(b), the changing trend of the frequencies with peak gains is similar to that of the resonance frequencies, but the peak gain values of the first and second resonant modes are insensitive to the variation of L , given by 8.5 and 10.5 dBi, respectively. For comparison, a DRA having a relatively small L of 28 mm is also simulated, and $l_s = 20$ mm, $l_m = 7$ mm are used for a good match. In this case, only the fundamental mode can be excited near 5 GHz. Its resonance frequency shifts up due to the decrease of DRA size. Therefore, a relatively large length to height ratio is very important for achieving a wide bandwidth.

Fig. 7 shows results for different heights (h_1) of the top dielectric layer. With reference to Fig. 7(a), when h_1 increases

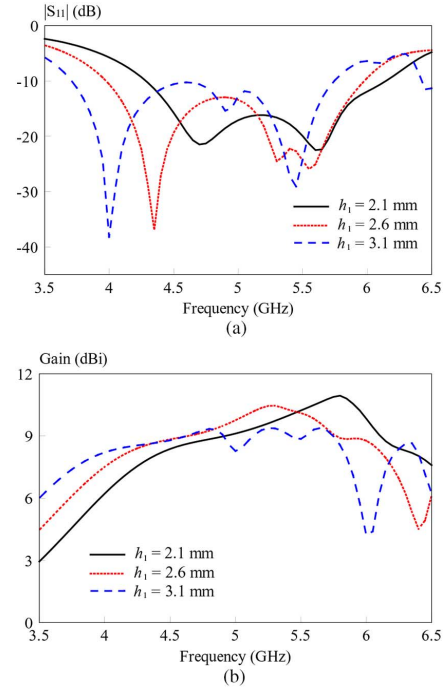


Fig. 7. Simulated reflection coefficients and gains of the wideband, low profile and high gain DRA as a function of frequency for different heights (h_1) of the top dielectric layer. (a) Reflection coefficients. (b) Antenna gains.

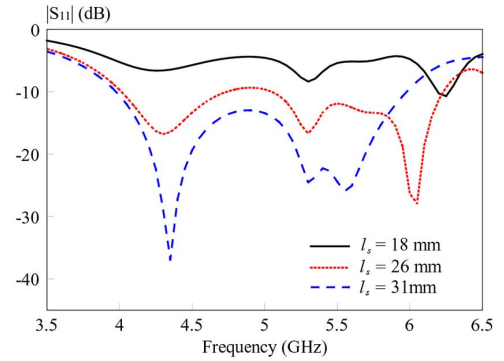


Fig. 8. Simulated reflection coefficients of the wideband, low profile and high gain DRA as a function of frequency for different slot lengths l_s .

from 2.1 to 3.1 mm, the lower resonance frequency decreases significantly from 4.7 to 4.0 GHz, whereas the upper resonance frequency remains almost unchanged at ~ 5.5 GHz. The results again verify that the lower and upper modes are due to the lateral and longitudinal resonances, respectively. It means that the two modes can be tuned separately, which is very desirable for the antenna design. Moreover, it can be observed from Fig. 7(b) that the variation of h_1 affects the gain level considerably. Therefore, choosing an appropriate height is very essential for realizing the high gain performance. Similar phenomena have been observed by varying the height (h_2) of the bottom dielectric layer, and thus they are not shown here for brevity.

Next, the effect of feeding slot is investigated. Fig. 8 shows the reflection coefficients for different slot lengths l_s . It is found that there are three resonant modes in the passband. The first

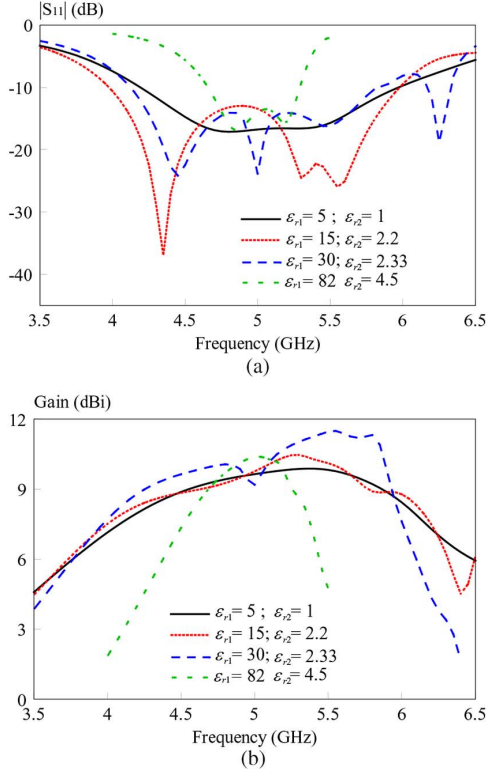


Fig. 9. Simulated reflection coefficients and gains of the wideband, low profile and high gain DRAs using different permittivities. (a) Reflection coefficients. (b) Antenna gains.

TABLE II
ANTENNA DIMENSIONS, BANDWIDTHS AND AVERAGE GAINS OF ANTENNAS I, II, III, AND IV

Antenna	Top layer $\epsilon_{r1} L \times L \times h_1$ (mm)	Bottom layer $\epsilon_{r2} L \times L \times h_2$ (mm)	slot size $l_s \times w_s$ (mm)	stub length l_m (mm)	Impedance Bandwidth	Average gain (dBi)
I	5, 48 × 48 × 5.8	1, 48 × 48 × 3.3	38 × 2	9.8	34.8 %	9.29
II	15, 46 × 46 × 2.6	2.2, 46 × 46 × 3	31 × 1.8	9	40.4 %	9.24
III	30, 48 × 48 × 1.4	2.33, 48 × 48 × 3.175	36 × 2	6.4	34.4 %	10.24
IV	82, 55 × 55 × 0.74	4.5, 55 × 55 × 1.52	28 × 2.2	5.4	12.6 %	9.77

and second resonances are insensitive to the variation of l_s as expected, because they are verified to be caused by the DRA. However, the resonance frequency of the third mode moves downward as l_s increases from 18 to 31 mm, showing that the third mode is related with the resonance of the slot. The slot mode merges with the second mode of the DRA when $l_s = 31$ mm. The length of the microstrip-stub l_m is also studied, and it is observed that the resonance frequency remains almost unchanged but the match level is greatly affected by changing l_m . Thus, l_m can be used to tune the match.

Finally, to demonstrate the flexibility of the present design, other three antennas using different combinations of dielectric layers have been designed to operate at ~ 5 GHz. In each case, the dimensions of dielectric layers and feeding slot are tuned to optimize the impedance match and gain level. Fig. 9 shows the simulated reflection coefficients and antenna gains for the three antennas. For ease of comparison, the results of the above prototype are also included in the figure. With reference to Fig. 9(a), the antenna with $\epsilon_{r1} = 5$ and $\epsilon_{r2} = 1$ has an impedance bandwidth of 34.8%. Similar bandwidth is obtained for the antenna with $\epsilon_{r1} = 30$ and $\epsilon_{r2} = 2.33$, but it significantly reduces to 12.6% when very high permittivities of

$\epsilon_{r1} = 82$ and $\epsilon_{r2} = 4$ are used. This is reasonable because the Q-factor increases with permittivity. It is worth mentioning the bandwidth of 12.6% is still ~ 4 times wider than the typical value of 3% for the very high permittivity design [12], [13]. With reference to Fig. 9(b), an average gain of higher than 9 dBi can always be obtained, and the antenna with $\epsilon_{r1} = 30$ and $\epsilon_{r2} = 2.33$ can even achieve a high gain of 10.2 dBi. Table II summarizes the antenna dimensions, bandwidths, and average gains for the four cases.

V. CONCLUSION

A stacked DRA consisting of two dielectric layers with different permittivities has been investigated in this letter. It has been demonstrated that by enlarging the length-to-height ratio and properly choosing the permittivities for the two dielectric layers, it is easy to realize an antenna possessing low profile, wide bandwidth and high gain. For demonstration, a prototype with permittivities of 15 and 2.2 was designed to operate at 5 GHz. The antenna has a measured impedance bandwidth of $\sim 40\%$ and an average gain of ~ 9 dBi. The gain can be further enhanced to ~ 10 dBi by using higher dielectric permittivities of 30 and 2.33 for the two dielectric layers.

REFERENCES

- [1] A. Petosa, *Dielectric Resonator Antenna Handbook*. Norwood, MA, USA: Artech House, 2007.
- [2] Nasimuddin and K. P. Esselle, "A low profile compact microwave antenna with high gain and wide bandwidth," *IEEE Trans. Antennas Propag.*, vol. 55, no. 6, pp. 1880–1883, Jun. 2007.
- [3] T. A. Denidni, Y. Coulibaly, and H. Boutayeb, "Hybrid dielectric resonator antenna with circular mushroom-like structure for gain improvement," *IEEE Trans. Antennas Propag.*, vol. 57, no. 4, pp. 1043–1049, Apr. 2009.
- [4] A. Petosa and S. Thirakoune, "Rectangular dielectric resonator antennas with enhanced gain," *IEEE Trans. Antennas Propag.*, vol. 59, no. 4, pp. 1385–1389, Apr. 2011.
- [5] A. Perron, T. A. Denidni, and A. R. Sebak, "High-gain hybrid dielectric resonator antenna for millimeter-wave applications: Design and implementation," *IEEE Trans. Antennas Propag.*, vol. 57, no. 10, pp. 2882–2892, Oct. 2009.
- [6] A. A. Kishk, X. Zhang, A. W. Glisson, and D. Kajfez, "Numerical analysis of stacked dielectric resonator antennas excited by a coaxial probe for wideband applications," *IEEE Trans. Antennas Propag.*, vol. 51, no. 8, pp. 1996–2006, Aug. 2003.
- [7] A. Petosa, N. Simons, R. Siushansian, A. Ittipiboon, and M. Cuhaci, "Design and analysis of multisegment dielectric resonator antennas," *IEEE Trans. Antennas Propag.*, vol. 48, no. 5, pp. 738–742, May 2000.
- [8] X. S. Fang and K. W. Leung, "Design of wideband omnidirectional two-layer transparent hemispherical dielectric resonator antenna," *IEEE Trans. Antennas Propag.*, vol. 62, no. 10, pp. 5353–5357, Oct. 2014.
- [9] K. X. Wang and H. Wong, "A circularly polarized antenna by using rotated-stair dielectric resonator," *IEEE Antennas Wireless Propag. Lett.*, vol. 14, pp. 787–790, 2015.
- [10] Y. Hwang, Y. P. Zhang, K. M. Luk, and E. K. N. Yung, "Gain-enhanced miniaturized rectangular dielectric resonator antenna," *Electron Lett.*, vol. 33, no. 5, pp. 350–352, Feb. 1997.
- [11] K. P. Esselle, "A low-profile rectangular dielectric resonator antenna," *IEEE Trans. Antennas Propag.*, vol. 44, no. 9, pp. 1296–1297, Sep. 1996.
- [12] R. K. Mongia, A. Ittipiboon, and M. Cuhaci, "Low profile dielectric resonator antennas using a very high permittivity material," *Electron Lett.*, vol. 30, no. 17, pp. 1362–1363, Aug. 1994.
- [13] K. W. Leung, K. M. Luk, E. K. N. Yung, and S. Lai, "Characteristics of a low profile circular disk DR antenna with very high permittivity," *Electron Lett.*, vol. 31, no. 6, pp. 417–418, Mar. 1995.